CFM

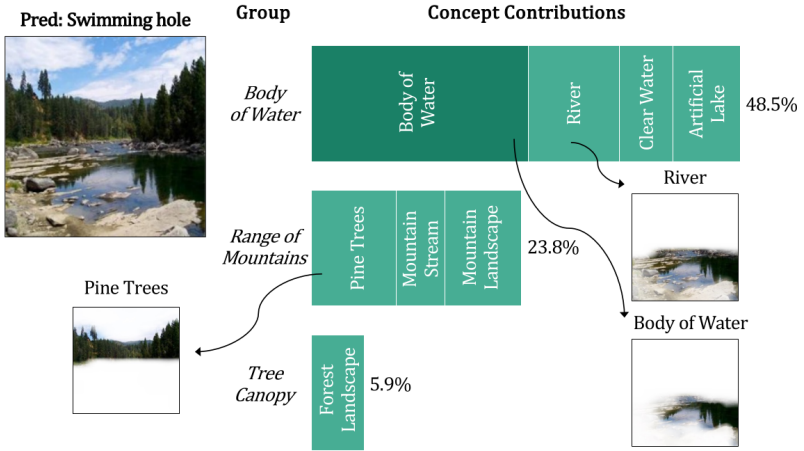
Language-aligned Concept Foundation Model for Vision

Julia Borowska & Mateusz Wątły

4 November 2026

The Problem: Explanations Can Be Misleading

Why do interpretable concepts need spatial grounding?



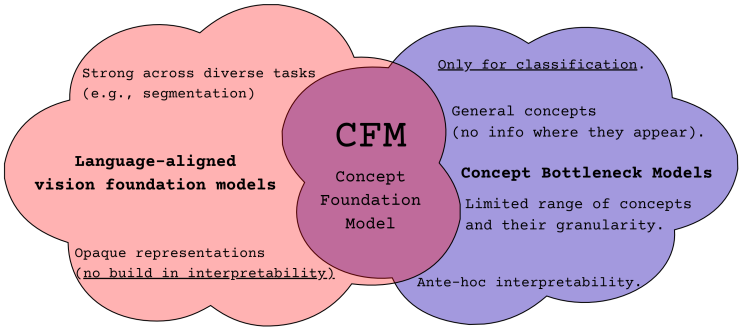
Other models: Absent concepts (*canoeing*, *swimsuits*).

CFM: Concepts grounded in image regions.

Fig. 6

Why Do We Need CFM?

Bridging foundation models and interpretable concepts



What Changes with CFM?

A reusable, spatially grounded concept representation

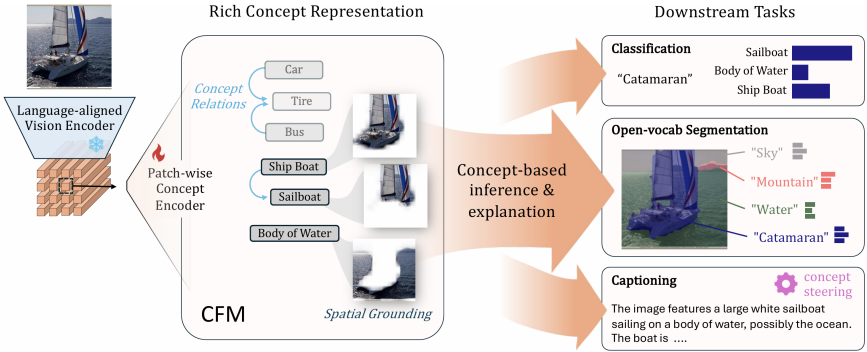


Fig. 1

Previous approaches

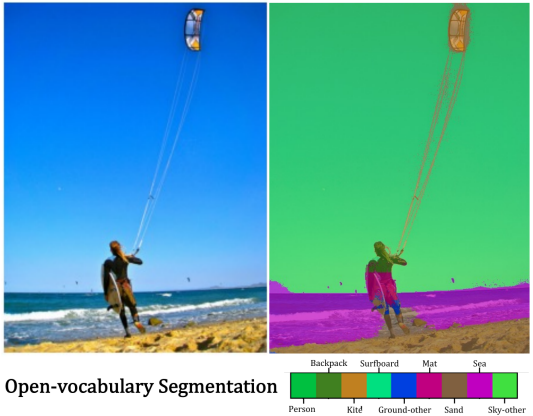
Mainly image-level concepts and classification-oriented explanations.

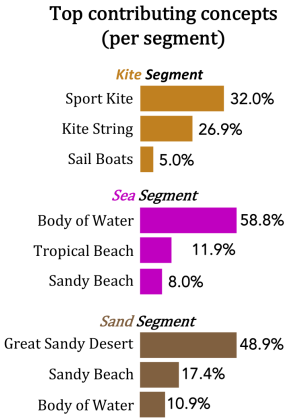
CFM

Spatially localized concepts with hierarchical relationships, reusable across different vision tasks.

Transparency of Model Predictions

Why does CFM matter for Trustworthy ML?





CFM provides:

- Localized predictions.
- Concepts contributing to each prediction.
- Concept contribution scores.

Trustworthy ML:

- Transparency
- Auditability
- Error analysis

Fig. 8

Interpretability facilitates model inspection, but does not guarantee correctness or robustness.

Contributions

CFM provides concepts that are:

- **Fine-grained:** at different levels of detail
- **Spatially grounded:** well localized in the image
- **Unified:** shared across different downstream tasks

Contributions

CFM provides concepts that are:

- **Fine-grained:** at different levels of detail
- **Spatially grounded:** well localized in the image
- **Unified:** shared across different downstream tasks

CFM automatically discovers concept relations:

- **Based on local co-occurrences:** concepts activated in the same image regions
- **Specifications:** e.g. *turtle* $\rightarrow$ *leatherback turtle*
- **Part-relations:** e.g. *turtle* $\rightarrow$ *turtle shell*

Contributions

CFM provides concepts that are:

- **Fine-grained:** at different levels of detail
- **Spatially grounded:** well localized in the image
- **Unified:** shared across different downstream tasks

CFM improves concept labeling by:

- **Matching concepts with text labels** from a large vocabulary
- Taking **concept hierarchies** into account
- Using **parent concepts** to better identify **specific concepts**

CFM automatically discovers concept relations:

- **Based on local co-occurrences:** concepts activated in the same image regions
- **Specifications:** e.g. *turtle* $\rightarrow$ *leatherback turtle*
- **Part-relations:** e.g. *turtle* $\rightarrow$ *turtle shell*

Contributions

CFM provides concepts that are:

- **Fine-grained:** at different levels of detail
- **Spatially grounded:** well localized in the image
- **Unified:** shared across different downstream tasks

CFM improves concept labeling by:

- **Matching concepts with text labels** from a large vocabulary
- Taking **concept hierarchies** into account
- Using **parent concepts** to better identify **specific concepts**

CFM automatically discovers concept relations:

- **Based on local co-occurrences:** concepts activated in the same image regions
- **Specifications:** e.g. *turtle* $\rightarrow$ *leatherback turtle*
- **Part-relations:** e.g. *turtle* $\rightarrow$ *turtle shell*

CFM provides:

- **Human-interpretable explanations**
- **Competitive performance** in classification, segmentation, and captioning
- **Steerable outputs** through concept representations

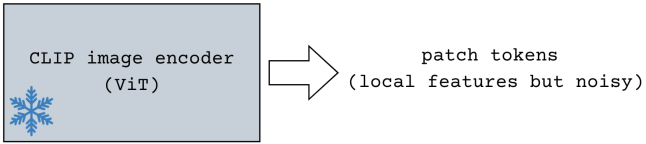
CFM Method

- Extracting Concept Representations
- Discovering Concept Relationships
- Hierarchy-aware Concept Labeling

Local representations from CLIP

GOAL

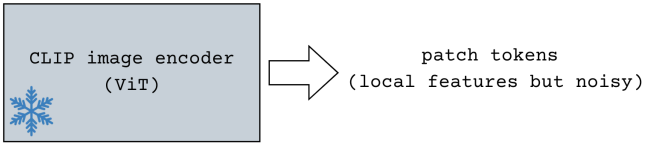
Semantically similar regions should have similar representations.

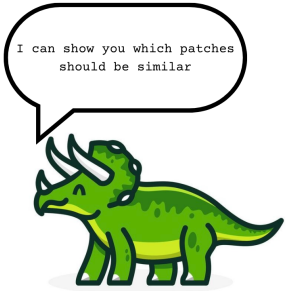


Local representations from CLIP

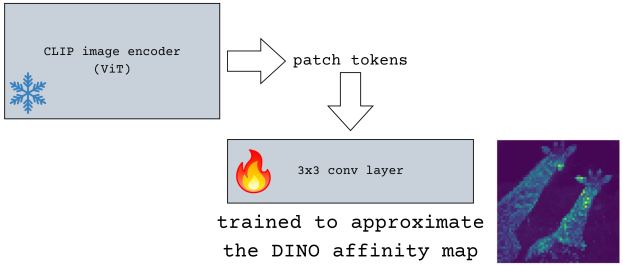
GOAL

Semantically similar regions should have similar representations.





Guided pooling



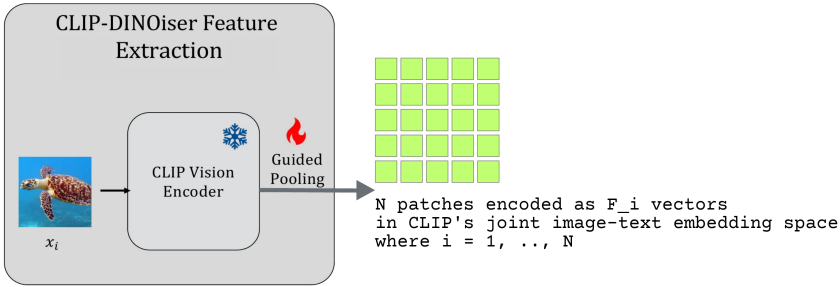
POOLING FORMULA

F_q: local feature of patch q
$A_{p,q}$: affinity between patches p and q

$$F_p^+ = \frac{\sum_{q=1}^{N} A_{p,q} F_q}{\sum_{q=1}^{N} A_{p,q}}$$

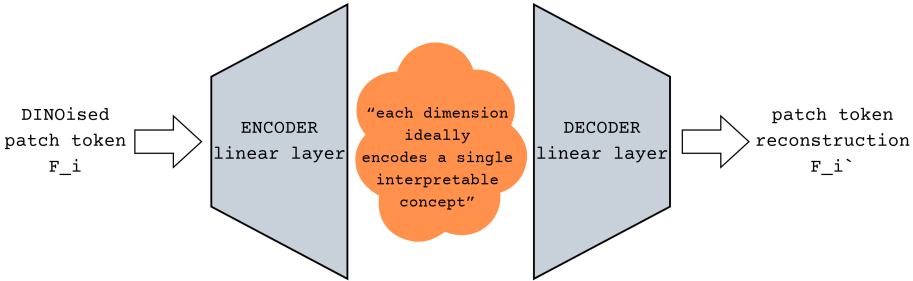
- Similar patches have a strong influence.
- Dissimilar patches contribute less or are ignored.

DINOised representations



- **Spatially localized** — each representation corresponds to an image patch.
- **Less noisy** — Guided Pooling reduces noise in local representations.
- **Semantically smoother** — visually similar regions can have more consistent features.
- **Language-aligned** — the representations retain a connection to CLIP's image-text embedding space.

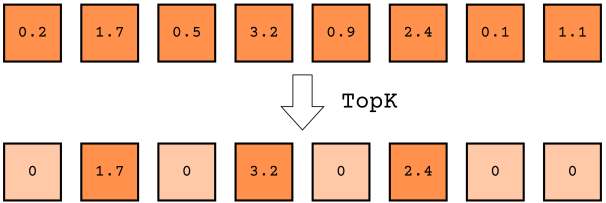
Sparse Autoencoder for concept extraction



TRAINING

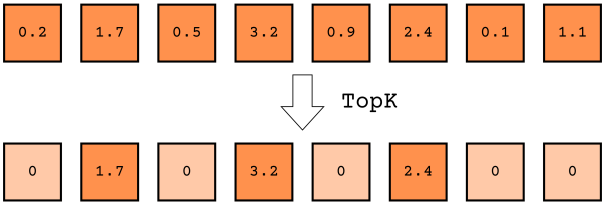
- reconstruction loss
- sparsity regularization term

BatchTopK: sparsity regularization



SAE decoder reconstructs the input using only the retained activations.

BatchTopK: sparsity regularization



SAE decoder reconstructs the input using only the retained activations.



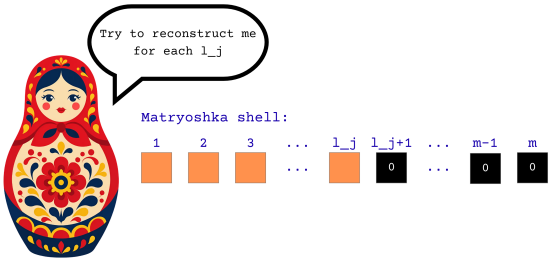
different patches may contain different number of concepts

BatchTopK

Matryoshka objective

F_p - DINOised patch token
$\pi^{enc} : \mathbb{R}^d \to \mathbb{R}^m$ - SAEs encoder
$\pi^{dec} : \mathbb{R}^m \to \mathbb{R}^d$ - SAEs decoder
l_j - from given set of indexes defining a matryoshka shell
(e.g., **l_j = [66, 311, 803, 1786, 3753, 8202]** for **m=8192**)

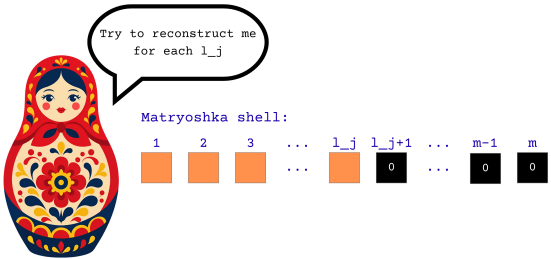
$$\mathcal{L}_{rec}(F_p) = \sum_j \left\| \pi^{dec}(\ \pi^{enc}(F_p)[1 : l_j]\ ) - F_p \right\|_2^2$$



Matryoshka objective

F_p - DINOised patch token
$\pi^{enc} : \mathbb{R}^d \to \mathbb{R}^m$ - SAEs encoder
$\pi^{dec} : \mathbb{R}^m \to \mathbb{R}^d$ - SAEs decoder
l_j - from given set of indexes defining a matryoshka shell
(e.g., `l_j = [66, 311, 803, 1786, 3753, 8202]` for `m=8192`)

$$\mathcal{L}_{rec}(F_p) = \sum_j \left\| \pi^{dec}(\ \pi^{enc}(F_p)[1 : l_j]\ ) - F_p \right\|_2^2$$

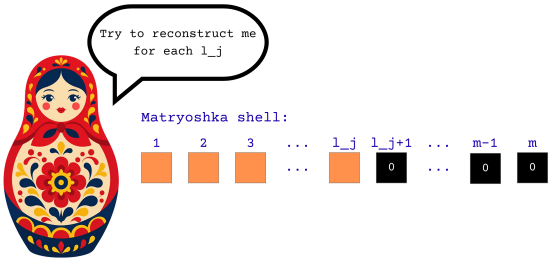


- Smaller parts should preserve useful information.
- *First neurons learn coarse, high-level information.*

Matryoshka objective

F_p - DINOised patch token
$\pi^{enc} : \mathbb{R}^d \to \mathbb{R}^m$ - SAEs encoder
$\pi^{dec} : \mathbb{R}^m \to \mathbb{R}^d$ - SAEs decoder
l_j - from given set of indexes defining a matryoshka shell
(e.g., `l_j = [66, 311, 803, 1786, 3753, 8202]` for `m=8192`)

$$\mathcal{L}_{rec}(F_p) = \sum_j \left\| \pi^{dec}(\ \pi^{enc}(F_p)[1 : l_j]\ ) - F_p \right\|_2^2$$



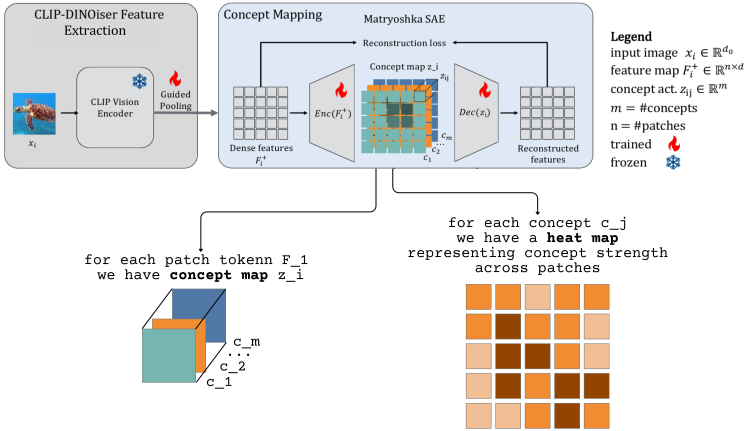
- Smaller parts should preserve useful information.
- *First neurons learn coarse, high-level information.*
- Larger parts can capture more information and finer distinctions.
- *Higher-index neurons learn the residual fine-grained information.*

From learned representations to named concepts

Learned representations:
localized concepts at diverse granularities.

Remaining challenge:
the dimensions in concept space
do not have human-readable names.

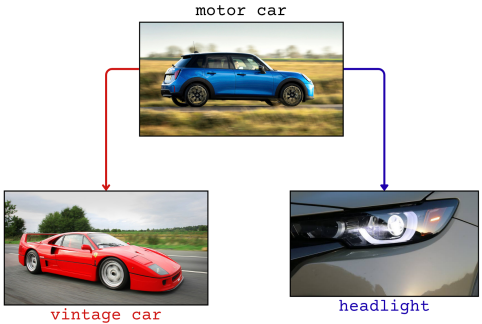
Next step:
identify concepts and assign labels.



Discovering inter-concept relationships

I. SPECIFICATIONS
PARENT: more general
CHILD: specialized

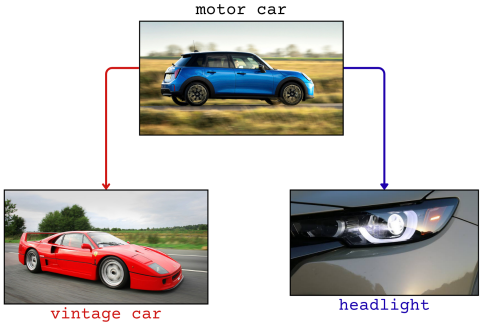
II. PART-RELATIONS
PARENT: whole object
CHILD: part of the object



Discovering inter-concept relationships

I. SPECIFICATIONS
PARENT: more general
CHILD: specialized

II. PART-RELATIONS
PARENT: whole object
CHILD: part of the object



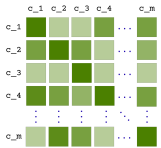
Hierarchy Criterion:

CHILD concept is active $\Longrightarrow$ PARENT concept should also be active

Measuring concept co-occurrence

Concept co-occurrence matrix: $C \in [0,1]^{m \times m}$

where C_{ij} is *conditional probability*
of concept c_i (potential PARENT) being active given the concept c_j (potential CHILD) is active



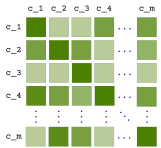
Calculating C_{ij}:

- A_{jp} - activation value of concept c_j for patch p

Measuring concept co-occurrence

Concept co-occurrence matrix: $C \in [0,1]^{m \times m}$

where C_{ij} is *conditional probability*
of concept c_i (potential PARENT) being active given the concept c_j (potential CHILD) is active



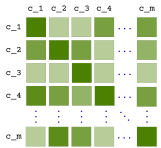
Calculating C_{ij}:

- A_{jp} - activation value of concept c_j for patch p
- $p : j \in B(p)$ - select all patches p for which the CHILD concept c_j is active

Measuring concept co-occurrence

Concept co-occurrence matrix: $C \in [0,1]^{m \times m}$

where C_{ij} is *conditional probability*
of concept c_i (potential PARENT) being active given the concept c_j (potential CHILD) is active



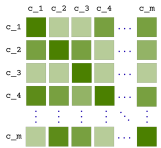
Calculating C_{ij}:

- A_{jp} - activation value of concept c_j for patch p
- $p : j \in B(p)$ - select all patches p for which the CHILD concept c_j is active
- $\sum_{p:j \in B(p)} A_{jp}$ - sum the activations of the CHILD concept c_j across the selected patches

Measuring concept co-occurrence

Concept co-occurrence matrix: $C \in [0,1]^{m \times m}$

where C_{ij} is *conditional probability*
of concept c_i (potential PARENT) being active given the concept c_j (potential CHILD) is active



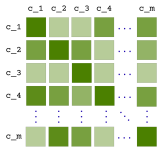
Calculating C_{ij}:

- A_{jp} - activation value of concept c_j for patch p
- $p : j \in B(p)$ - select all patches p for which the CHILD concept c_j is active
- $\sum_{p:j \in B(p)} A_{jp}$ - sum the activations of the CHILD concept c_j across the selected patches
- $p : j, i \in B(p)$ - select all patches p for which both the PARENT concept c_i and the CHILD concept c_j are active

Measuring concept co-occurrence

Concept co-occurrence matrix: $C \in [0,1]^{m \times m}$

where C_{ij} is *conditional probability*
of concept c_i (potential PARENT) being active given the concept c_j (potential CHILD) is active



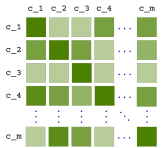
Calculating C_{ij}:

- A_{jp} - activation value of concept c_j for patch p
- $p : j \in B(p)$ - select all patches p for which the CHILD concept c_j is active
- $\sum_{p:j \in B(p)} A_{jp}$ - sum the activations of the CHILD concept c_j across the selected patches
- $p : j, i \in B(p)$ - select all patches p for which both the PARENT concept c_i and the CHILD concept c_j are active
- $\sum_{p:j,i \in B(p)} A_{jp}$ - sum the activations of the CHILD concept c_j across the selected patches

Measuring concept co-occurrence

Concept co-occurrence matrix: $C \in [0,1]^{m \times m}$

where C_{ij} is *conditional probability* of concept c_i (potential PARENT) being active given the concept c_j (potential CHILD) is active



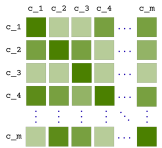
Calculating C_{ij}:

- A_{jp} - activation value of concept c_j for patch p
- $p : j \in B(p)$ - select all patches p for which the CHILD concept c_j is active
- $\sum_{p:j \in B(p)} A_{jp}$ - sum the activations of the CHILD concept c_j across the selected patches
- $p : j, i \in B(p)$ - select all patches p for which both the PARENT concept c_i and the CHILD concept c_j are active
- $\sum_{p:j,i \in B(p)} A_{jp}$ - sum the activations of the CHILD concept c_j across the selected patches
- $C_{ij} = \frac{\sum_{p:j,i \in B(p)} A_{jp}}{\sum_{p:j \in B(p)} A_{jp}}$

Measuring concept co-occurrence

Concept co-occurrence matrix: $C \in [0,1]^{m \times m}$

where C_{ij} is *conditional probability* of concept c_i (potential PARENT) being active given the concept c_j (potential CHILD) is active



Do we have PARENT $c_i \rightarrow$ CHILD c_j relationship?

for a chosen threshold τ (e.g., $\tau = 0.75$)

- YES - if $C_{ij} \geq \tau$
- NO - otherwise

Calculating C_{ij}:

- A_{jp} - activation value of concept c_j for patch p
- $p : j \in B(p)$ - select all patches p for which the CHILD concept c_j is active
- $\sum_{p:j \in B(p)} A_{jp}$ - sum the activations of the CHILD concept c_j across the selected patches
- $p : j, i \in B(p)$ - select all patches p for which both the PARENT concept c_i and the CHILD concept c_j are active
- $\sum_{p:j,i \in B(p)} A_{jp}$ - sum the activations of the CHILD concept c_j across the selected patches
- $C_{ij} = \frac{\sum_{p:j,i \in B(p)} A_{jp}}{\sum_{p:j \in B(p)} A_{jp}}$

From concepts to language

Concept dictionary vector π_i^{dec}

$\pi^{dec} : \mathbb{R}^m \to \mathbb{R}^d$ - SAE decoder affine map
$W^{dec} \in \mathbb{R}^{d \times m}$ - decoder weight matrix
b^{dec} - decoder bias vector
$e_i \in \mathbb{R}^m$ - one-hot vector corresponding to concept c_i

$$\begin{aligned} \pi^{dec}(e_i) &= W^{dec} e_i + b^{dec} \\ &= \boldsymbol{W_{:,i}^{dec}} + b^{dec} \\ &= \boldsymbol{\pi_i^{dec}} + b^{dec} \end{aligned}$$

The concept dictionary vector for concept c_i is the i**-th column** of the decoder weight matrix:

$$\pi_i^{dec} = W_{:,i}^{dec}$$

From concepts to language

Concept dictionary vector π_i^{dec}

$\pi^{dec} : \mathbb{R}^m \to \mathbb{R}^d$ - SAE decoder affine map
$W^{dec} \in \mathbb{R}^{d \times m}$ - decoder weight matrix
b^{dec} - decoder bias vector
$e_i \in \mathbb{R}^m$ - one-hot vector corresponding to concept c_i

$$\begin{aligned} \pi^{dec}(e_i) &= W^{dec} e_i + b^{dec} \\ &= \boldsymbol{W^{dec}_{:,i}} + b^{dec} \\ &= \boldsymbol{\pi_i^{dec}} + b^{dec} \end{aligned}$$

The concept dictionary vector for concept c_i is the i**-th column** of the decoder weight matrix:

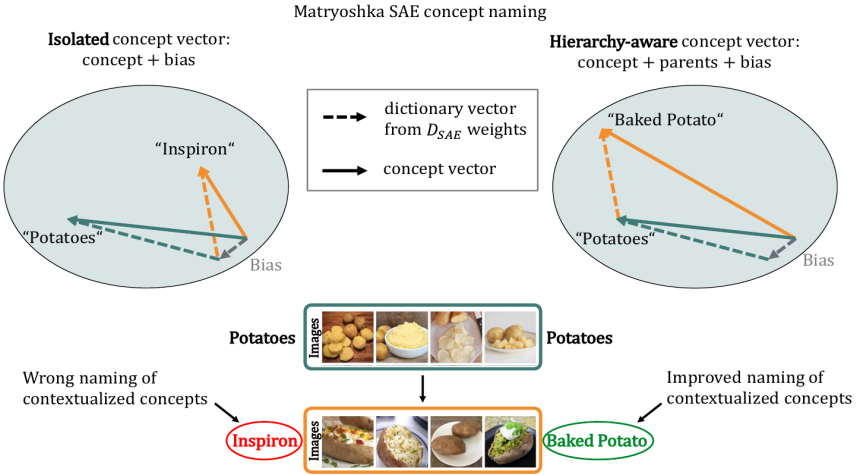
$$\pi_i^{dec} = W^{dec}_{:,i}$$

Standard Approach

$$l_i = \arg\max_{v \in \mathcal{V}} \cos\left(\pi_i^{dec}, \mathcal{T}(v)\right)$$

- $\mathcal{V}$ - pre-defined vocabulary of candidate labels
- $\mathcal{T}(v) \in \mathbb{R}^d$ - CLIP text embedding of $v \in \mathcal{V}$
- l_i - label assigned to concept c_i

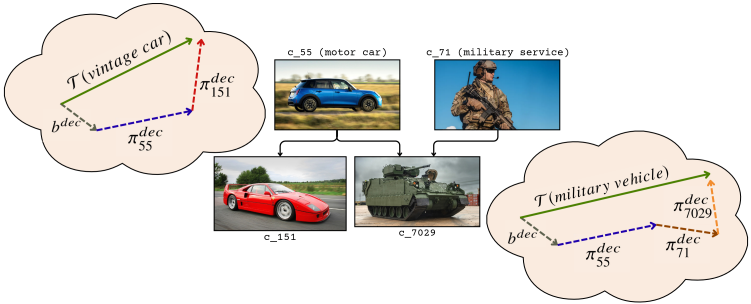
Taking parents into account can help



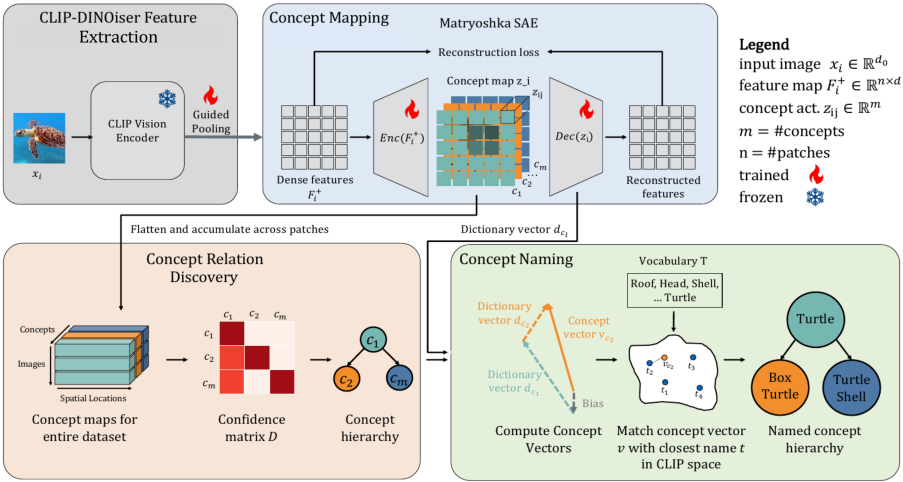
Hierarchy-aware concept naming

$$l_j = \arg\max_{v \in \mathcal{V}} \cos\left(\left(\sum_{(i \to j) \in G} \pi_i^{dec}\right) + \pi_j^{dec} + b^{dec}, \mathcal{T}(v)\right)$$

- G - concept hierarchy graph
- $i \to j$ - concept c_i is a parent of concept c_j
- π_k^{dec} - concept dictionary vector of c_k
- b^{dec} - decoder bias vector
- $\mathcal{T}(v)$ - CLIP text embedding of candidate label v
- $\mathcal{V}$ - predefined vocabulary of candidate labels
- l_j - label assigned to concept c_j



CFM: from images to interpretable concepts



CFM Key Results

- Improved Concept Quality and Naming
- Performance Across Downstream Tasks
- Steering through Concept-Level Intervention

Improved Concept Quality

Comparison with previous approaches

Concept quality benchmarks

Method	Locality ↑		Consistency ↑		Impurity ↓	
	Part	Stuff	Part	Stuff	Part	Stuff
SALF-CBM	16.4	19.5	9.9	6.7	.841	1.143
PatchSAE	32.2	34.6	13.1	12.7	.353	.663
CFM	**44.3**	**44.5**	**15.3**	**13.1**	**.333**	**.539**

Part = PartImageNet; Stuff = COCO-Stuff.

Main finding

CFM learns concepts that are more spatially localized, more consistent and less impure than competing methods.

Human Evaluation

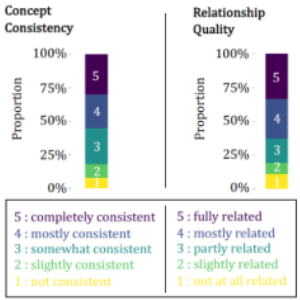


Fig. 3

Human ratings of concept consistency and relationship quality.

Concept Naming Accuracy

Do the assigned labels describe what is actually visible?

+42.8

percentage points

Improvement in image-level concept-naming accuracy over a random baseline.

Other methods

DN-CBM: +20.4 pp
MSAE: +26.2 pp

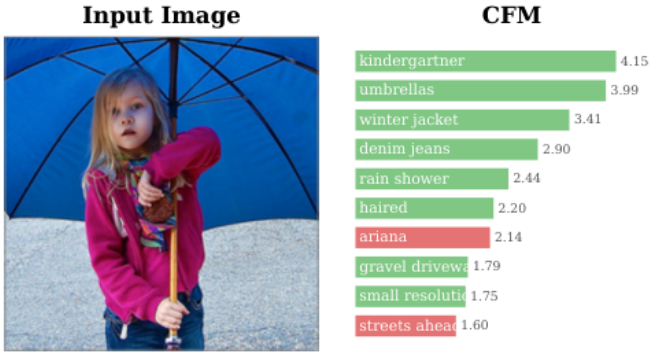


Fig. 5: Green labels match visible content; red labels do not.

Evaluation: top-10 concept names per image on COCO, judged by GPT-5-mini.

Performance Across Downstream Tasks

What is the performance cost of interpretability?

Task and metric	Opaque baseline	CFM
ImageNet, top-1 accuracy (%)	80.2	78.9
Places365, top-1 accuracy (%)	55.1	**55.4**
Open-vocabulary segmentation, mean mIoU	41.4	41.3
Image captioning, CLIP-Score	0.72	0.72

Main finding

CFM retains competitive performance while adding spatially grounded, interpretable concepts.

CLIP linear probe is the classification baseline; CLIP-DINOiser is used for segmentation and captioning. Fig. 6, Table 1, and Fig. 7.

Steerable Image Captioning

Intervening on named concepts changes the generated caption

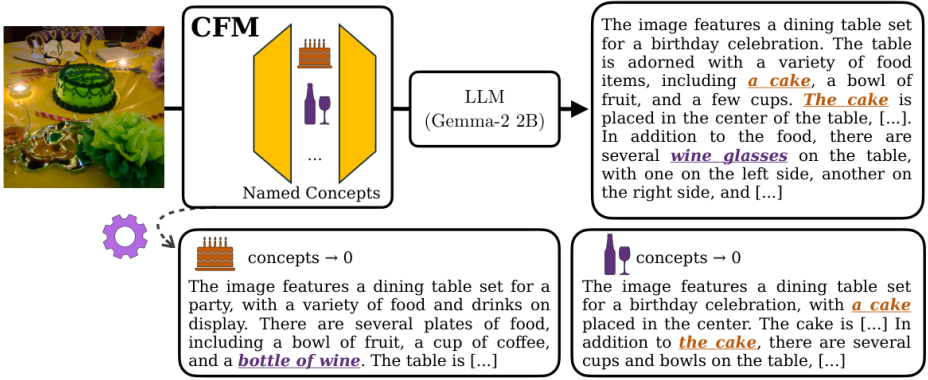


Fig. 7

Suppressing the *cake* concept removes it from the caption, while suppressing the *wine-glass* concept preserves the cake description.

Critical Limitations

Concept names are still imperfect

Some labels remain inaccurate or overly specific, even after hierarchy-aware naming.

Critical Limitations

Concept names are still imperfect

Some labels remain inaccurate or overly specific, even after hierarchy-aware naming.

Interpretability does not guarantee faithfulness

Spatially plausible concepts and successful interventions do not establish that every explanation captures the full causal basis of a prediction.

Critical Limitations

Concept names are still imperfect

Some labels remain inaccurate or overly specific, even after hierarchy-aware naming.

Interpretability does not guarantee faithfulness

Spatially plausible concepts and successful interventions do not establish that every explanation captures the full causal basis of a prediction.

Generalization needs broader testing

The reported experiments cover selected backbones, datasets, and tasks. Reliability under stronger distribution shifts remains an open question.

Discussion

Question 1:

Could CFM become for interpretable computer vision what CLIP became for vision-language understanding or SAM for segmentation?

Discussion

Question 1:

Could CFM become for interpretable computer vision what CLIP became for vision-language understanding or SAM for segmentation?

Question 2:

Are co-activation patterns sufficient to establish meaningful parent–child relationships between concepts?